

Amit Singh

Senior Backend Engineer | Python, AWS & Production Reliability

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Software engineer with 7+ years across financial data, network operations, and product-safety systems. Build and operate Python services on AWS, with a focus on transactional correctness, cross-service debugging, and recovery from failure.

PROFESSIONAL EXPERIENCE

Realtek Software Services | Amazon Ring / Lab126 client engagement

Software Development Engineer II · 2025–present · Pune, India

- Own backend and platform delivery for a multi-company product-safety application, spanning Python services, SQLAlchemy data access, relational schema changes, and React operator interfaces.
- Traced a cross-service authorization regression to an identity-resolution change in a separately owned permission service; drove the correction at its source instead of adding a local workaround.
- Designed compliance-sensitive reconciliation across UI changes, asynchronous Jira events, and scheduled repair to recover from missed updates; implemented company-aware behaviour across database and API boundaries.
- Restored a critical missing-data workflow within approximately 10 hours, then strengthened database-change controls with reviewed SQL, rollback plans, staged rehearsal, and post-change verification.
- Migrated Claude inference after model deprecation and added confidence-based human review to text/image analysis. Mentor two engineers and participate in the on-call rotation.

Orange Business Services

Senior Software Engineer · Project-based engagement · 2024–2025

- Built and productionized Jenkins/Jira/Groovy pipelines for multi-vendor device integration and configuration updates, supporting beta, user-acceptance, and production rollout.
- Delivered a ServiceNow-grounded operations assistant using OpenSearch, embeddings, and an LLM, with human verification for uncertain responses and a weekly knowledge-refresh lifecycle.
- Led user testing and feedback-driven refinement; carried services through integration testing, runbooks, customer guides, and operational handover.

Virtusa Polaris | Lloyds Banking Group engagement

Software Engineer · 2019–2024

- Built APIs and ETL pipelines integrating collateral/risk data and market feeds for financial instruments within regional delivery windows.
- Led PostgreSQL migration and Snowflake-integrated data pipelines; improved API performance through Python optimization and SQL tuning.
- Monitored production pipelines with Dynatrace and handled incidents through ServiceNow, maintaining operational continuity across data dependencies.

SELECTED INDEPENDENT PROJECTS

CloudScale Backend | Python, PostgreSQL

Built an event-sourced ledger with transaction-bound idempotency, balanced postings, and a transactional outbox. Added PostgreSQL integration tests and an executable backup/restore drill that rebuilds derived state. Recovery evidence is at seeded scale; sustained availability remains unproven.

FastAPI Microservices Platform | Python, PostgreSQL

Built self-hosted webhook delivery without Redis or Kafka, using lease-fenced workers, Standard Webhooks signatures, and isolated egress. A reproducible million-job correctness run recorded no lost jobs; delivery remains at-least-once and the backlog-latency target was not met.

Design decisions, benchmark scope, and pinned CI evidence: amitsinghom.github.io/evidence.html. Further work includes an offline code analyzer and durable agent workflows.

TECHNICAL SKILLS

Backend: Python, FastAPI, Django, SQLAlchemy, REST APIs, event-driven services, idempotency, transactional data processing.

Cloud & operations: AWS Lambda, API Gateway, Step Functions, SQS/SNS, IAM, CloudWatch, Docker, Kubernetes, CI/CD, incident response.

Data: SQL, PostgreSQL, Aurora MySQL, DynamoDB, Snowflake, schema design, migrations, ETL.

Supporting: React, TypeScript, Amazon Bedrock, retrieval-grounded AI, OpenSearch; Go through hands-on projects.

EDUCATION

Bachelor of Engineering, Information Technology | University of Mumbai